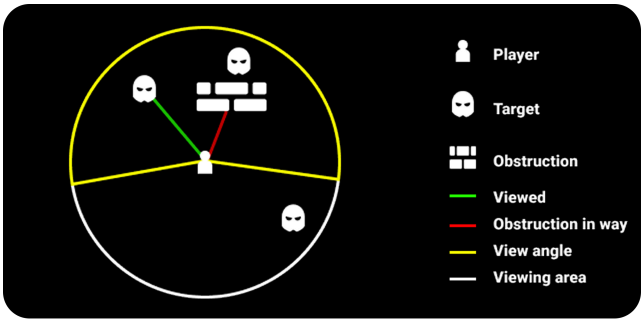
# Field Of View (FOV)

**Overview** The **"FieldOfView"** is designed to create a field of view for detecting objects within a specified radius and angle, It's particularly useful for implementing vision or detection systems in Unity projects.




### Features

- **Field Parameters:** Allows customization of the field's radius, angle and offset from the origin.
- ***Target and Obstruction Masks*:** Specify the layers to be considered as targets and obstructions for detection.
- ***Field Method*:** Includes a method `Field<T>(string tag = null)` that returns a list of objects of a specified type (`T`) within the defined field of view based on certain conditions.

### Usage

- **Attach the Script:** Attach the `"FieldOfView.cs"` script to a GameObject in your Unity scene.
- **Set Parameters:** Adjust the radius, angle, offset, and layer masks in the Inspector according to your requirements.
- **Call the Field Method:** Use the `Field<T>()` method to retrieve a list of objects within the defined field of view.

### Example Usage:

```
// Example usage in another script
List<YourObjectType> detectedObjects = GetComponent<FieldOfView>().Field<YourObjectType>("YourTag");
// Replace YourObjectType with the actual type of object you're looking for and "YourTag" with the specific
// tag condition (or leave it null to ignore tag conditions).
```

### Notes:

- *Ensure the necessary colliders and tags are appropriately set up on the objects you want to detect within the field of view.*
- *Make sure to handle the visibility checks and responses to detected objects as per your game or application requirements.*